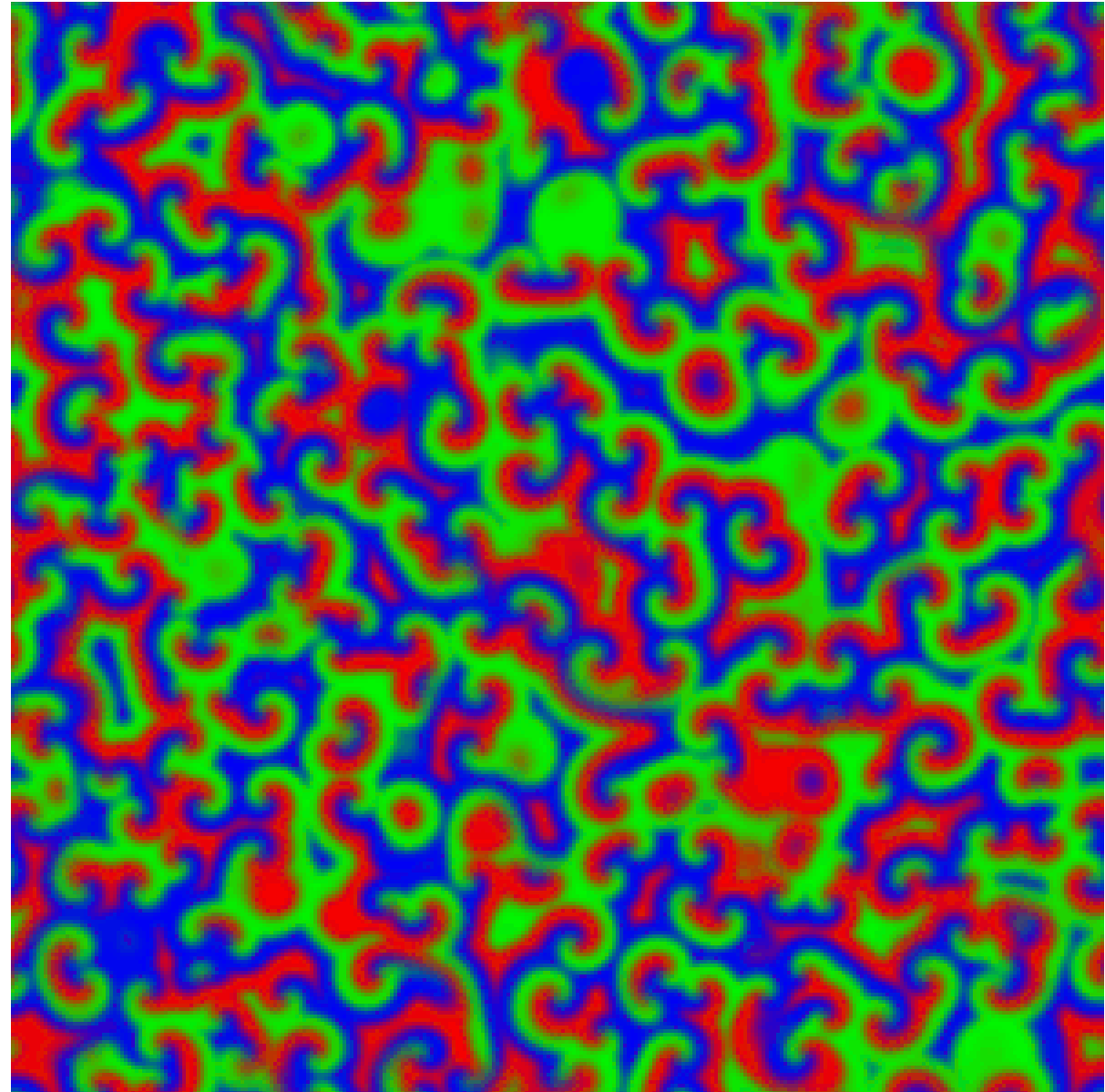


Day 21 - Oscillations

Belousov–Zhabotinsky reaction →

An example of spatiotemporal
patterns



Announcements

- Midterm 1 is being graded
- Midterm 1 solutions are posted
- Homework 5 is due on Friday (late on Sunday)
 - Anyone may request an extension for HW5 up to one week, no questions asked. Just email me.
- Homework 6 is due the following Friday (late on Sunday)
 - Again, anyone may request an extension for HW6 up to one week, no questions asked. Just email me.
- Class next Monday, maybe?

Clicker Question 21-1

My 15 year old asked me to ask you this question.

She has a physics test next week. 🙏🙏🙏

Just a vibe check. How are the vibes?

1. The vibes are immaculate.
2. The vibes are good.
3. The vibes are okay.
4. The vibes are not good.
5. The vibes are terrible.

Reminders

We were solving nonlinear first order differential equations of the homogenous type

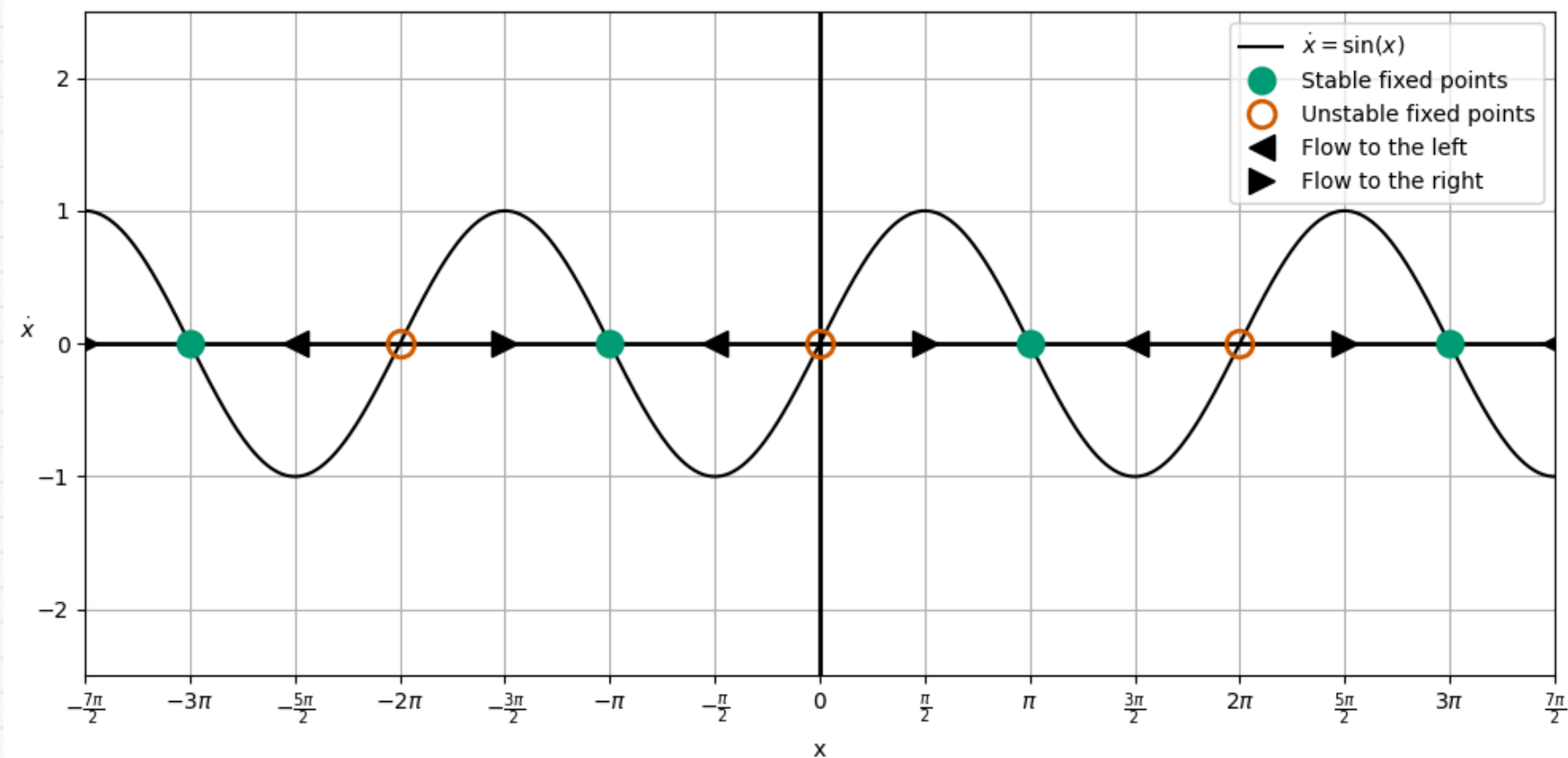
$$\frac{dx}{dt} + f(x) = 0$$

where $f(x)$ is a nonlinear function of x . We found critical points where $dx/dt = 0$.

We demonstrated the utility of phase diagrams to visualize the behavior of solutions to nonlinear differential equations.

Reminders

$$\dot{x} = \sin x$$

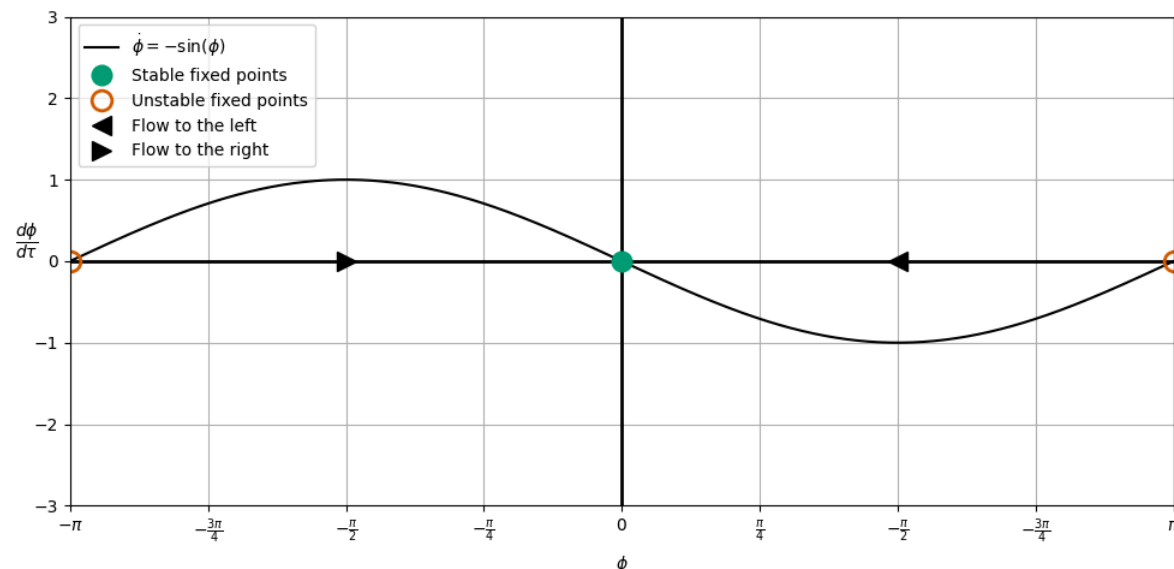


Reminders

We showed that we can learn something about systems we know little about at first using this approach (i.e., Firefly synchronization).

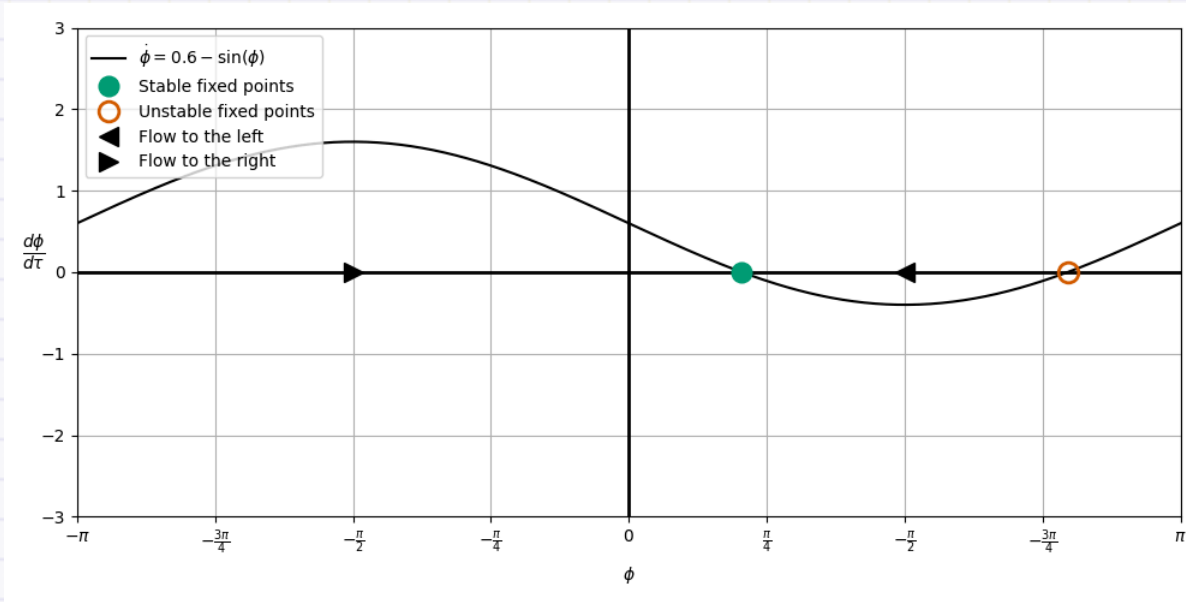
$$\frac{d\phi}{d\tau} = \mu - \sin(\phi)$$

$\mu = 0 \longrightarrow$ Synchronization always (no phase difference)



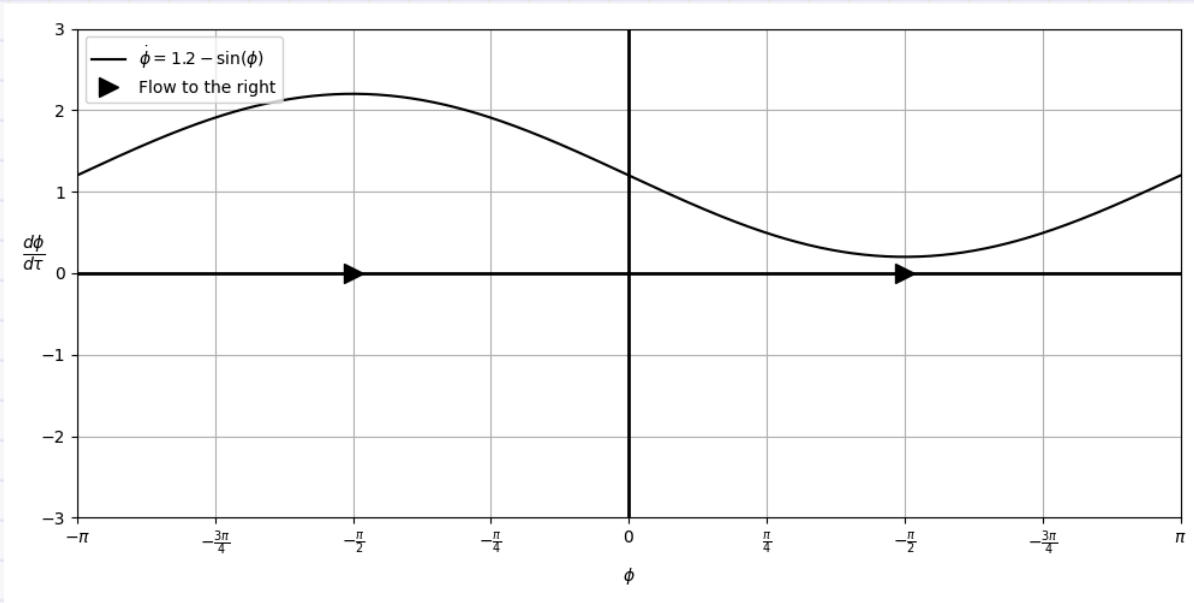
Reminders

$\mu = 0.6 \longrightarrow$ Entrainment is possible (constant phase difference)



Reminders

$\mu = 1.2 \longrightarrow$ No entrainment ($\Omega > \omega$). Stimulus is too fast.



Reminders

We then moved on to 2D phase spaces, where we had a system of two first order equations:

$$\begin{aligned}\dot{x} &= v \\ \dot{v} &= f(x, v)\end{aligned}$$

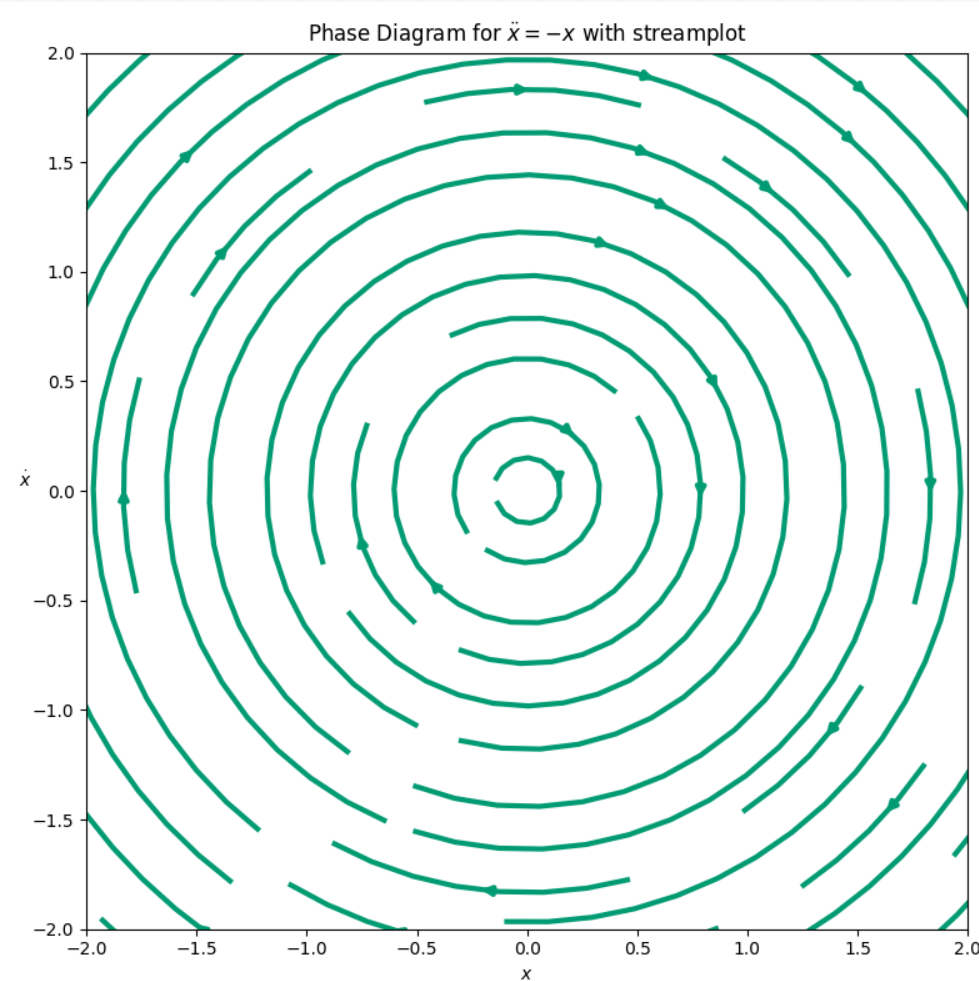
We focused on the simple harmonic oscillator as an example.

$$\begin{aligned}\ddot{x} &= -x \\ \dot{x} &= v\end{aligned}$$

And we graphed it's phase space diagram.

Reminders

Phase space diagram for a simple harmonic oscillator. The ellipses are curves of constant energy, E .



$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E$$
$$\frac{v^2}{2E/m} + \frac{x^2}{2E/k} = 1$$

Large Angle Pendulum

In the case of a large angle pendulum, we have a nonlinear differential equation:

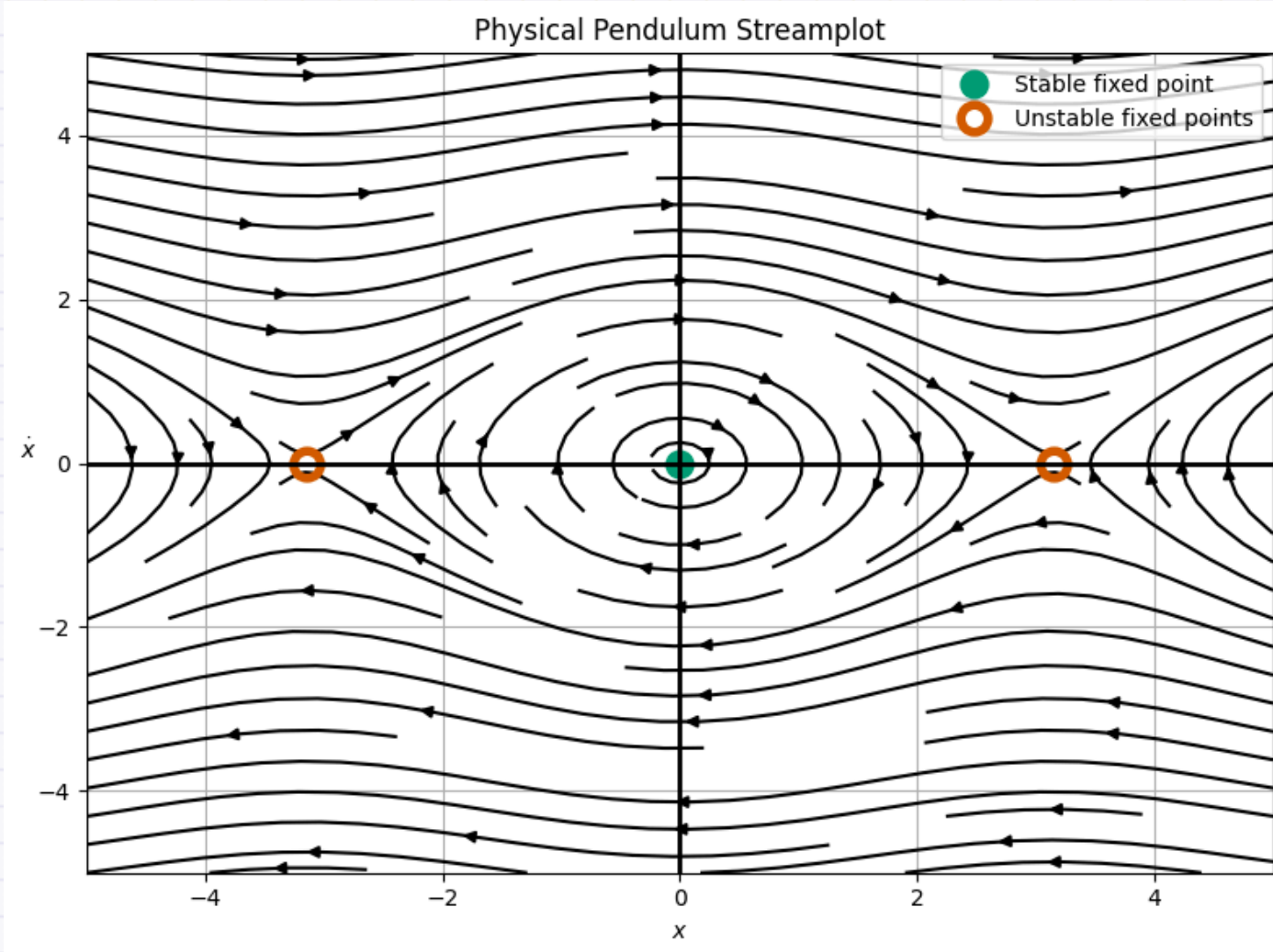
$$\ddot{\theta} = -\frac{g}{L}\sin \theta$$

Here, we can write this as a system of two first order equations:

$$\begin{aligned}\dot{\theta} &= \omega \\ \dot{\omega} &= -\frac{g}{L}\sin \theta\end{aligned}$$

We can then plot the phase space diagram for this system.

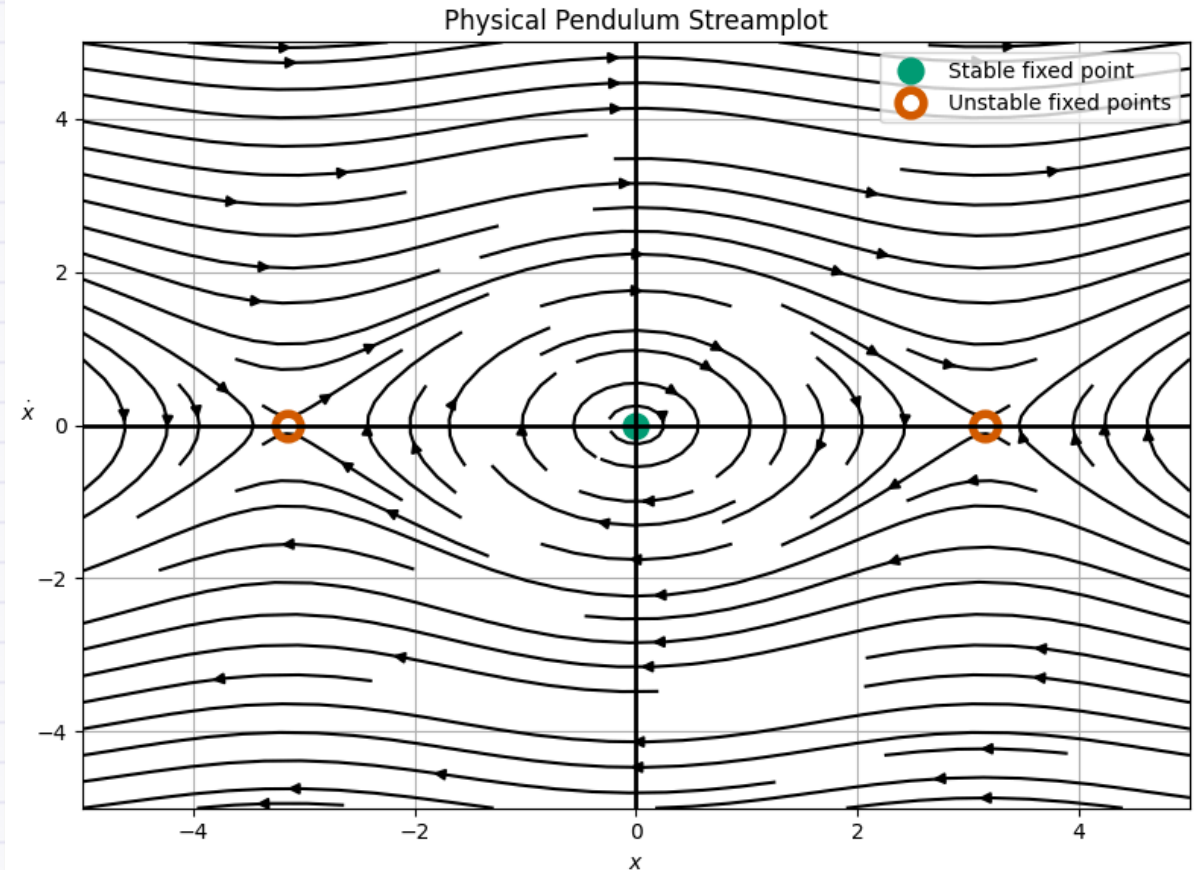
Phase Diagram for a Large Angle Pendulum



Clicker Question 21-2

Consider the phase diagram for a large angle pendulum. What do the upper and lower flows represent?

1. Unrealistic solutions.
2. Solutions that are not physical.
3. Solutions that are physical.
4. Solutions well beyond small angle.
5. More than one of these.



Oscillations

Table Discussion: Why do we see oscillations so frequently in nature?

- Can you think of at least two different reasons from different perspectives?
- What are some non-simple harmonic oscillators you can think of?

Click when you and your table are done.

Clicker Question 21-3

Consider the following notation:

$$z = x + iy$$

$$z^* = x - iy$$

$$zz^* = x^2 + y^2$$

Is this an idea you've seen before? How do you feel about it?

1. Yes, seen it and comfortable with it.
2. Yes, seen it but not comfortable with it.
3. No, never seen it, but I can learn it.

There is no option for not learning, sorry.

Visualizing the Complex Solution

We constructed a solution of the form:

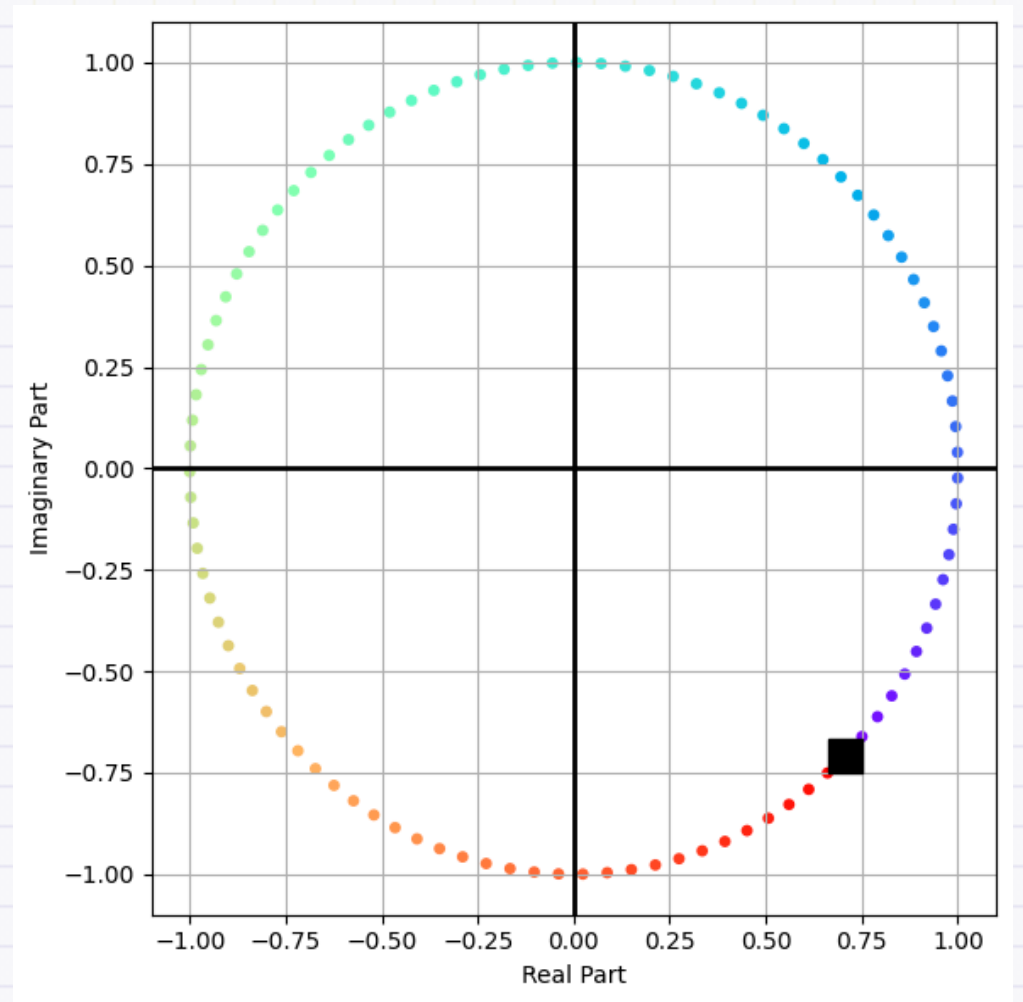
$$Ae^{i\omega t - \delta}$$

We can plot it in the complex plane and see the real and imaginary parts, and how they change in time.

Visualizing the Complex Solution

We can plot the solution on the complex plane. For this, $\delta = \pi/4$, and the amplitude is $A = 1$.

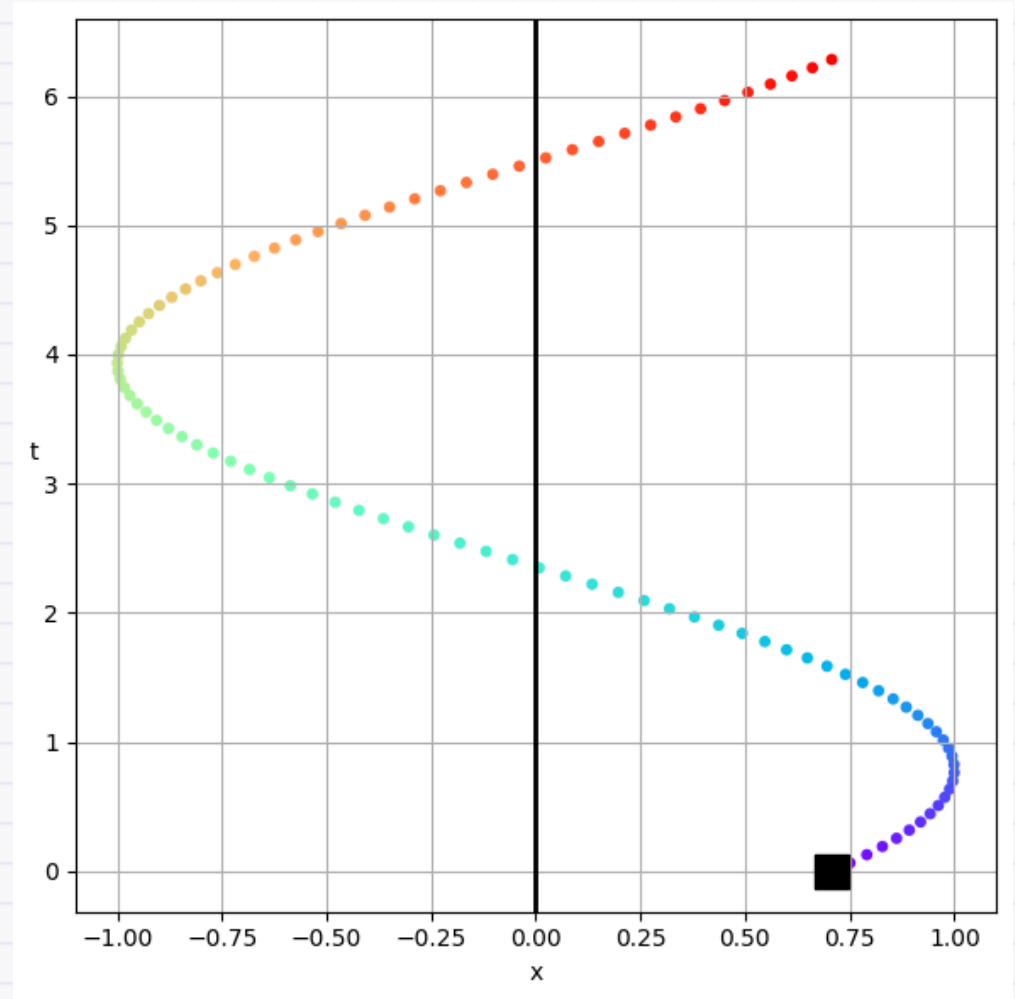
The solution rotates counterclockwise in the complex plane, following the rainbow from violet to red.



Projecting the Real Solution

The real part is just the projection of the complex solution onto the real axis. Just how far along the real axis is the solution at any given time.

That looks like a time trace, but not quite, it's the real projection. The colors scheme is the same as before.



The Time Trace of the Solution

We just flip the axes to produce the time trace that you are used to seeing. The color scheme is the same as before.

